

Tool Name		Scaphandre	Keple ≥ 0.10	Kepler ≤ v0.9	Green Metric Tool	e-Surgeon (PowerAPI)	BitWatts	PowerJoular	Joular Core	e-Surgeon (eIalén)	JoularIX	lgProf (Energy module)	JRAPL
Granularity	Virtual Machine- Pod-Level	X	X	X	no	no	no	X	X	no	no	X	no
	Container-Level	X	X	X	X	no	no	no	no	no	no	no	no
	Application-Level	X	X	X	no	no	no	no	no	no	no	no	no
	Process-Level	X	X	X	no	X	X	X	X	no	no	X	no
	Thread-Level	no	no	no	no	no	X	X	no	X	no	X	no
	Code Section-Level	no	no	no	X(SCI functionality)	no	no	no	no	X	X	X	X
Measurement Sources	CPU	Hardware-Level Energy Estimation	no	no	no	X(Compatible with certain Metric Providers)	no	no	no	no	no	no	no
			no	X (RAPL)	X (RAPL)	X (RAPL)	no	no	X (RAPL through powercap)	X (RAPL (linux/windows) or powerMetrics (Mac))	no	X (RAPL through powercap)	X (RAPL through linux MSR kernel module)
		Model-based	no	X Regression-based models for VM deployment	X Regression-based models for VM deployment	X1) XGBoost model (TDP, Machine Specs and CPUs) 2) Analytical Model(TDP, #CPUchips and CPU %)	X Linear Regression Model (HPC: "Instructions", "cache references", "cache misses")	X HPC vector (cycles, instructions, cache misses)	no	X (TDP) Through PowerAPI	no	no	no
			no	OS-provided PID CPU % Cgroups Stat	PID runtime + HPC vector (cycles, instructions, cache misses) through eBPF tracing	OS-provided PID CPU % Cgroups Stat Code Special Print (Code Section)	OS-provided PID CPU %	OS-provided PID CPU %	OS-provided PID CPU %	OS-provided PID CPU %	1) Time Markers : JVM byte code instrumentation / 2) Call stack	1) OS-provided PID CPU % for Java Threads through JVM instrumentation 2) Call Stack	Call stack Time Markers
	DRAM	Hardware-Level Energy Estimation	no	no	no	no	no	no	no	no	no	no	no
			no	X (RAPL)	X (RAPL)	X (RAPL)	no	no	X (RAPL)	X (RAPL)	no	X (RAPL) Through Power Joular	X (RAPL) X (RAPL through linux MSR kernel module)
		Model-based	no	no	no	no	no	no	no	no	no	no	no
			no	OS-provided PID CPU % Cgroups Stat	PID runtime + HPC vector (cycles, instructions, cache misses) through eBPF tracing	OS-provided PID CPU % Cgroups Stat (Container) Code Special Print (Code Section)	no	no	PID CPU %	PID CPU %	no	1) OS-provided PID CPU % for Java Threads through JVM instrumentation 2) Statistical sampling of thread stack trace	Call stack sampling Time Markers
	GPU	Hardware-Level Energy Estimation	no	no	no	no	no	no	no	no	no	no	no
			no	X (NVLM)	X (NVLM)	X (NVLM)	no	no	X (NVIDIA-SMI, AMD-SMI) (computers and servers)	X (NVIDIA-SMI, AMD-SMI) (computers and servers)	no	X (NVIDIA SMI) (computers and servers through Power Joular)	no
		Model-based	no	no	no	no	no	no	no	no	no	no	no
			no	NVLM-provided (i.e. driver) per-process SM utilization	NVLM-provided (i.e. driver) per-process SM utilization	OS-provided PID CPU % Cgroups Stat Code Special Print (Code Section)	no	no	PID GPU % (from driver API)	PID GPU % (from driver API)	no	1) OS-provided PID CPU % for Java Threads through JVM instrumentation 2) Statistical sampling of thread stack trace	no
	Network (NIC)	Hardware-Level Energy Estimation	no	no	no	no	no	no	no	no	no	no	no
			no	no	no	no	no	no	no	no	no	no	no
		Model-based	no	no	no	no	no	no	no	no	no	no	no
			no	no	no	no	X	no	no	no	X (Through PowerAPI)	no	no
	System	Hardware-Level Energy Estimation	no	no	no	no	no	no	no	no	no	no	no
			X (RAPL)	no	no	X (RAPL PSYS Domain)	no	no	X (RAPL)	no	no	X (RAPL through PAPI)	X (RAPL through linux MSR kernel module)
		Model-based	no	Regression-based model	Regression-based trained power model	X1) XGBoost model (TDP, Machine Specs and CPUs) 2) Analytical Model(TDP, #CPUchips and CPU %)	X (TDP)	X HPC (cycles, instructions, cache misses) vector	polynomial regression For Raspberry Pi	polynomial regression For Raspberry Pi	no	polynomial regression For Raspberry Pi	no
			OS-provided PID CPU %	OS-provided PID CPU % Cgroups Stat	PID runtime + HPC vector (cycles, instructions, cache misses) through eBPF tracing	OS-provided PID CPU % Cgroups Stat (container) Code Special Print (Code Section)	OS-provided PID CPU %	OS-provided PID CPU %	OS-provided PID CPU %	OS-provided PID CPU %	no	1) OS-provided PID CPU % for Java Threads through JVM instrumentation 2) Statistical sampling of thread stack trace	Call stack sampling Time Markers
Allocation Model	Activity Share Estimation	Directly	X	X	X	X	X	X	X	X (For NIC)	X (for threads)	no	no
		Proxy Reconstruction	no	no	X (from HPC vector (cycles, instructions, cache misses))	no	no	no	no	no	no	no	no
		Statistical Sampling	no	no	no	no	no	no	no	X (for CPU : call-stack sampling version)	X	X (1) Allocate all RAPL system energy to process it is attached to 2) Statistical Sampling (stack trace) to allocate energy to functions	no
	Execution Time Markers	Execution Time Markers	no	no	no	X (SCI functionality)	no	no	no	X (for CPU : bytecode instrumentation version)	no	no	X
		Baseline Modelling and Calibration	no	Modeled and allocated proportionnaly to processes size	Modeled and allocated proportionnaly to processes size	Modeled and allocated to container according to OS-provided PID CPU %	no	no	no	Modeled and subtracted before allocation	no	no	no
Virtualised environment support	Pretrained Model	Pretrained Model	no	Regression model (# CPU and OS-provided Resource Utilisation metrics as input)	Regression model (# CPU and OS-provided Resource Utilisation metrics as input)	1) XGBoost model (TDP, Machine Specs and CPUs) 2) Analytical Model(TDP, #CPUchips and CPU %)	no	no	no	no	no	no	no
		Two instances : host exposing VM-level metrics to guest through shared file	X	no ((CNCf 2023) mentions a planned "passthrough" mode for VM deployment which expose VM energy in a shared file.	no ((CNCf 2023) mentions a planned "passthrough" mode for VM deployment which expose VM energy in a shared file.	no	no	no	X	no	no	no	no
	Delegated to another	Delegated to another	no	no	no	no	no	no	no	no	Uses PowerJoular	no	no